

A Sharp Interior Power-Zero Threshold for the One-Dimensional Free Poincaré Inequality

Candidate partial result for arXiv:1105.2031

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Status: candidate partial result, likely valid. The source's broad regularity question is not claimed solved in full.

1 Source question and normalization

Michel Ledoux and Ionel Popescu ask in Remark 2 of *The One Dimensional Free Poincaré Inequality* [1] (arXiv:1105.2031, PDF page 15):

Poincaré's inequality and the C^2 condition on the density w imply that w must be positive. Also, if w is positive everywhere and continuous then $P(\rho)$ holds for some ρ . It is interesting to see what happens if the C^2 condition on w is dropped. Is it still true that there is a Poincaré inequality satisfied for some $\rho > 0$? And if so, under what are the regularity conditions on w ?

$$\sum_{n=1}^{\infty} \lambda_n^{-\alpha} = \sum_{n=1}^{\infty} \lambda_n^{-\alpha} \lambda_n$$

which is obviously true. Written in terms of the operator $\mathcal{N}$, (2.2) is equivalent to

$$\text{Var}_{\beta}(f) \leq \langle \mathcal{N}f, f \rangle$$

for all $f \in C^2([-2, 2])$. Or this is just the spectral gap of $\mathcal{N}$. Equality is attained in (2.2) only for f linear. The free Poincaré's is actually equivalent to the statement $\mathcal{N} \leq \mathcal{N}^2$. As $\mathcal{N}$ is a non-negative operator, this is in fact equivalent to (2.2).

(4) Follows from $P(1/2)$ for α . □

Remark 2. Poincaré's inequality and the C^2 condition on the density w imply that w must be positive. Also, if w is positive everywhere and continuous then $P(\rho)$ holds for some ρ . It is interesting to see what happens if the C^2 condition on w is dropped. Is it still true that there is a Poincaré inequality satisfied for some $\rho > 0$? And if so, under what are the regularity conditions on w ?

Figure 1: Remark 2 in arXiv:1105.2031, PDF page 15.

We use the source normalization on $[-2, 2]$. Set

$$\alpha(dx) = \frac{\sqrt{4-x^2}}{2\pi} dx, \quad \beta(dx) = \frac{dx}{\pi\sqrt{4-x^2}},$$

and

$$\omega(dx dy) = \frac{4-xy}{4\pi^2\sqrt{(4-x^2)(4-y^2)}} dx dy.$$

For a smooth real function f , write

$$\mathcal{Q}(f) = \iint \left(\frac{f(x) - f(y)}{x - y} \right)^2 \omega(dx dy), \quad \mathcal{E}_w(f) = \int (f')^2 w d\alpha.$$

A probability measure $\mu = w\alpha$ satisfies $P(\rho)$ when

$$2\rho \mathcal{Q}(f) \leq \mathcal{E}_w(f) \tag{1}$$

for every smooth f on $[-2, 2]$.

For $0 < \theta < \pi$, put

$$x = 2 \cos \theta, \quad W(\theta) = w(2 \cos \theta).$$

Thus β becomes $d\theta/\pi$, while α becomes $(2/\pi) \sin^2 \theta d\theta$.

2 Proof intuition

The source diagonalizes the free energy with Chebyshev polynomials. If $g = \mathcal{U}f$ and

$$k(\theta) = g(2 \cos \theta) \sin \theta,$$

then k is a sine series and $\mathcal{Q}(f)$ is exactly the weighted coefficient sum $\sum n|b_n|^2$. This borderline one-dimensional $H^{1/2}$ quantity controls every finite L^p norm of k . Consequently, Hölder's inequality controls $\int g^2/w d\alpha$ by $\mathcal{Q}(f)$ whenever W^{-1} has slightly more than L^1 integrability. A weighted Cauchy–Schwarz inequality then gives (1), with no differentiability assumption on w .

The reverse mechanism is weighted capacity. Near an interior zero of order $\gamma > 1$, a transition across a window of width ε costs only $O(\varepsilon^{\gamma-1})$ in $\mathcal{E}_w$. At the borderline $\gamma = 1$, the optimal logarithmic transition costs $O(1/\log(1/\varepsilon))$. These functions converge to a nonconstant step. The source estimate $\mathcal{Q}(f) \geq \frac{1}{2} \text{Var}_\beta(f)$ keeps their free energy bounded away from zero, so (1) cannot hold. The two mechanisms meet at the sharp exponent $\gamma = 1$.

3 A rough-density sufficient condition

Let $\phi_n(x) = T_n(x/2)$ and $\psi_n(x) = U_n(x/2)$ be the Chebyshev systems used in [1]. The operator

$$(\mathcal{U}f)(x) = \int \frac{f(x) - f(y)}{x - y} \beta(dy)$$

satisfies $\mathcal{U}\phi_n = \frac{1}{2}\psi_{n-1}$ for $n \geq 1$.

Proposition 1 (Reciprocal-integrability criterion). *Let $\mu = w\alpha$ be a probability measure, with $w > 0$ almost everywhere. If*

$$W^{-1} \in L^r(0, \pi) \quad \text{for some } r > 1, \tag{2}$$

then μ satisfies $P(\rho)$ for some $\rho > 0$. No continuity or differentiability of w is required.

Proof. It is enough first to consider a polynomial f ; the final inequality then extends to smooth f by its Chebyshev expansion. Write

$$g = \mathcal{U}f = \sum_{m \geq 0} b_m \psi_m.$$

The identities in [1] give

$$\mathcal{Q}(f) = \langle (\mathcal{M} + I)g, g \rangle_{L^2(\alpha)} = \sum_{m \geq 0} (m+1)|b_m|^2, \quad (3)$$

where $\mathcal{M}\psi_m = m\psi_m$. Since

$$\psi_m(2 \cos \theta) = \frac{\sin((m+1)\theta)}{\sin \theta},$$

we have

$$k(\theta) := g(2 \cos \theta) \sin \theta = \sum_{m \geq 0} b_m \sin((m+1)\theta). \quad (4)$$

Fix $2 \leq p < \infty$ and set $q = p/(p-1) \in (1, 2]$. The Hausdorff–Young inequality for the odd extension of (4), followed by Hölder’s inequality for sequences, gives

$$\begin{aligned} \|k\|_{L^p(0, \pi)} &\leq C_p \|(b_m)\|_{\ell^q}, \\ \sum_{m \geq 0} |b_m|^q &\leq \left(\sum_{m \geq 0} (m+1)|b_m|^2 \right)^{q/2} \left(\sum_{m \geq 0} (m+1)^{-q/(2-q)} \right)^{(2-q)/2}. \end{aligned}$$

The last series converges because $q > 1$ (the case $q = 2$ is immediate). Therefore

$$\|k\|_{L^p(0, \pi)}^2 \leq C_p' \mathcal{Q}(f) \quad (5)$$

for every finite $p \geq 2$.

Let $r' = r/(r-1)$ and take $p = 2r'$. From the change of variables above, (2), Hölder’s inequality, and (5),

$$\begin{aligned} J(f) &:= \int \frac{g^2}{w} d\alpha = \frac{2}{\pi} \int_0^\pi \frac{|k(\theta)|^2}{W(\theta)} d\theta \\ &\leq \frac{2}{\pi} \|W^{-1}\|_{L^r} \|k\|_{L^{2r'}}^2 \leq C \mathcal{Q}(f). \end{aligned} \quad (6)$$

Finally, the source identity

$$\mathcal{Q}(f) = \int f' \mathcal{U}f d\alpha$$

and weighted Cauchy–Schwarz yield

$$\mathcal{Q}(f)^2 \leq \left(\int (f')^2 w d\alpha \right) \left(\int \frac{(\mathcal{U}f)^2}{w} d\alpha \right) \leq C \mathcal{E}_w(f) \mathcal{Q}(f).$$

After cancelling $\mathcal{Q}(f)$ (the zero case is trivial), we obtain $\mathcal{E}_w(f) \geq C^{-1} \mathcal{Q}(f)$. Thus (1) holds with $\rho = (2C)^{-1} > 0$. $\square$

4 The sharp power-zero threshold

Theorem 2 (Finite interior power zeros). *Let $\mu = w\alpha$ be a probability measure. Assume there are finitely many points $a_1, \dots, a_N \in (-2, 2)$ and exponents $\gamma_j \geq 0$ such that, in pairwise disjoint neighborhoods of the a_j ,*

$$c_j |x - a_j|^{\gamma_j} \leq w(x) \leq C_j |x - a_j|^{\gamma_j}, \quad (7)$$

with $c_j, C_j > 0$, and assume w is bounded above and bounded away from zero off those neighborhoods. Then

$$\mu \text{ satisfies } P(\rho) \text{ for some } \rho > 0 \iff \max_j \gamma_j < 1. \quad (8)$$

Proof. Suppose first that $\gamma_* := \max_j \gamma_j < 1$. Choose $r > 1$ with $r\gamma_* < 1$. Since $x = 2 \cos \theta$ is bi-Lipschitz near every interior point a_j , (7) implies $W(\theta)^{-r}$ is locally integrable near each corresponding θ_j . Away from those points, W^{-r} is bounded. Thus (2) holds, and Proposition 1 proves $P(\rho)$.

Conversely, suppose that (7) holds at an a_j with exponent $\gamma := \gamma_j \geq 1$, and write $a_j = 2 \cos \theta_0$. For $h(\theta) = f(2 \cos \theta)$, a direct change of variables gives

$$\mathcal{E}_w(f) = \frac{1}{2\pi} \int_0^\pi W(\theta) |h'(\theta)|^2 d\theta, \quad \text{Var}_\beta(f) = \text{Var}_{d\theta/\pi}(h). \quad (9)$$

If $\gamma > 1$, take a fixed smooth transition from 0 to 1 and compress it into a window of width $O(\varepsilon)$ around θ_0 . The upper bound in (7) and (9) give

$$\mathcal{E}_w(f_\varepsilon) = O(\varepsilon^{\gamma-1}) \longrightarrow 0. \quad (10)$$

If $\gamma = 1$, use on one side of θ_0 the logarithmic profile

$$h_\varepsilon(\theta_0 + t) = \frac{\log(t/\varepsilon)}{\log(\delta/\varepsilon)}, \quad \varepsilon \leq t \leq \delta, \quad (11)$$

extended by 0 before the transition and by 1 after it. Smooth the two corners inside comparable subintervals. Since $|h'_\varepsilon| = (t \log(\delta/\varepsilon))^{-1}$ on the main transition, (7) and (9) give

$$\mathcal{E}_w(f_\varepsilon) \leq \frac{C}{\log^2(\delta/\varepsilon)} \int_\varepsilon^\delta \frac{dt}{t} = O\left(\frac{1}{\log(\delta/\varepsilon)}\right) \longrightarrow 0. \quad (12)$$

The smoothing can be chosen with the same bound.

In both cases, h_ε converges in $L^2(d\theta)$ to the nonconstant step $\mathbf{1}_{(\theta_0, \pi)}$ (up to reversing orientation). Consequently

$$\text{Var}_\beta(f_\varepsilon) \longrightarrow \frac{\theta_0}{\pi} \left(1 - \frac{\theta_0}{\pi}\right) > 0.$$

Proposition 3.2(3) of [1] states

$$\mathcal{Q}(f) \geq \frac{1}{2} \text{Var}_\beta(f). \quad (13)$$

Thus $\mathcal{Q}(f_\varepsilon)$ stays bounded away from zero while $\mathcal{E}_w(f_\varepsilon) \rightarrow 0$, contradicting (1) for every $\rho > 0$. This proves the reverse implication in (8). $\square$

Corollary 3 (Exact one-zero family). *Fix $a \in (-2, 2)$ and $\gamma \geq 0$, and normalize*

$$w_{a,\gamma}(x) = \frac{|x - a|^\gamma}{\int_{-2}^2 |t - a|^\gamma \alpha(dt)}.$$

Then $w_{a,\gamma}\alpha$ satisfies a free Poincaré inequality if and only if $\gamma < 1$. In particular, dropping C^2 does not preserve the inequality automatically: the continuous full-support density $w_{a,1}$ is a counterexample. On the other hand, every $0 < \gamma < 1$ gives a density which vanishes at a but still satisfies $P(\rho)$ for some $\rho > 0$.

5 Verification, scope, and novelty check

The normalization, the identities for $\mathcal{U}$, $\mathcal{M}$, and $\mathcal{Q}$, and the lower bound (13) were checked directly against [1]. The exponent computation in the sufficiency proof is sharp at the method’s endpoint: with $p = 2r'$ one has $q = p/(p - 1) = 2r/(r + 1)$ and $q/(2 - q) = r > 1$. The necessity proof separately checks the power and logarithmic transition costs.

This packet gives a complete characterization for finitely many interior power-law zeros and a more general sufficient condition. It does not characterize arbitrary measurable or oscillatory densities, endpoint degeneracies, singular components, or measures with several support intervals. For that reason it is a partial result rather than a claimed full answer to the source’s broad regularity question.

A bounded literature search checked arXiv:1105.2031; the exact wording of Remark 2; combinations of “free Poincaré”, “density regularity”, “power weight”, “semicircle”, “fractional Hardy”, and “interior zero”; and the later refinement paper of Houdré and Popescu [2] (arXiv:1311.4585). The latter develops higher-order and Brascamp–Lieb refinements, not the rough-density or power-zero question. No source stating the criterion or the exponent-one threshold above was located. This is bounded evidence, not a guarantee of novelty.

Human-review recommendation. Review as a likely valid substantial partial result. The highest-value checks are the Chebyshev normalization in (3), the finite- p sine-series estimate (5), and the approximation of the logarithmic profile in the borderline case $\gamma = 1$.

References

- [1] M. Ledoux and I. Popescu, *The One Dimensional Free Poincaré Inequality*, arXiv:1105.2031 (2011), <https://arxiv.org/abs/1105.2031>.
- [2] C. Houdré and I. Popescu, *Refinements of the One Dimensional Free Poincaré Inequality*, arXiv:1311.4585 (2013), <https://arxiv.org/abs/1311.4585>.
- [3] L. Grafakos, *Classical Fourier Analysis*, 3rd ed., Graduate Texts in Mathematics 249, Springer, 2014.